

Alyssa B. Sanchez

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EDUCATION

Colegio de Muntinlupa

Bachelor of Science in Computer Engineering

Sucat, Muntinlupa City

August 2021 – July 2025

- **Honors:** *cum laude*, GWA: 1.69
- **Relevant Coursework:** Object-Oriented Programming, Data Structures and Algorithms, Database Management Systems, Artificial Intelligence with Data Analytics, Operating Systems

EXPERIENCE

SBT Holdings Inc.

IT Intern

Alabang, Muntinlupa City

June 2024 – August 2024

- Developed a warehouse-inspired inventory and request management system that digitized manual tool tracking and implemented **Role-Based Access Control (RBAC)** using **Laravel**, **MySQL**, and **MVC architecture**.
- Designed a relational database schema and implemented **SQL-based queries** using **Laravel Eloquent ORM** to manage inventory records, user roles, and request transactions, ensuring data integrity and efficient retrieval.
- Participated in iterative development workflows, gathering requirements, designing wireframes and database schemas, and conducting regular progress reviews with supervisors to continuously improve system functionality.
- Utilized **Git** and **GitHub Actions CI/CD** pipelines to automatically run tests on code pushes, helping detect integration issues early during team collaboration.
- Recreated complex website layouts achieving full cross-device responsiveness using **Tailwind CSS**, **HTML**, and **JavaScript** to adapt UI components across mobile and desktop devices.
- Gained exposure to enterprise IT infrastructure by assisting in server room audits and performing routine hardware maintenance and troubleshooting.

PROJECTS

AsteraCare: AIoT-Based Smart Flower Preservation System | *Kotlin, YOLOv8, ESP32* | [GitHub]

2025

Project Lead & App Developer

- Led the conceptualization and system architecture of an AIoT-based flower preservation system, designing integration across mobile app, AI model, microcontroller, and environmental hardware components.
- Developed a native Android app using **Kotlin** and **XML**, serving as the primary monitoring and control interface of the system.
- Architected the end-to-end data flow from on-device AI classification to hardware execution, ensuring low-latency communication between the Android interface and microcontroller sensors.
- Integrated a pre-trained **YOLOv8 AI** model optimized with **TensorFlow Lite (TFLite)** into the app, enabling real-time flower species identification to trigger automated, species-specific preservation logic.
- Implemented **Bluetooth SPP** protocols with an **ESP32** microcontroller to facilitate seamless hardware-to-app communication, allowing bidirectional real-time sensor data streaming and actuator control.
- Optimized application performance by utilizing **multithreading** and **lifecycle-aware components**, ensuring 100% UI stability during high-frequency sensor data streaming and AI inference.

Movie Finder Web App | *React.js, Tailwind CSS, TMDB API, Netlify* | [GitHub]

2025

- Built a responsive movie search web application using **React.js**, **Tailwind CSS**, and **TMDB API**, gaining hands-on experience with component-based architecture, API integration, and state management.
- Deployed a functional web app to practice end-to-end development, real-world workflows, and API data handling.

SKILLS

Programming: Python, SQL, HTML, CSS, JavaScript, Java, Kotlin, PHP, C/C++, C#, MATLAB, Verilog

Frameworks & Libraries: Laravel, Tailwind CSS, React.js, Bootstrap

Tools & Platforms: Git/GitHub, GitHub Actions, Canva, TinkerCAD, Proteus 8, Figma, SolidWorks, FluidSIM

CERTIFICATIONS

- Civil Service Honor Graduate Eligibility (P.D. 907) | 2025
- HCIA-Artificial Intelligence V3.5 (Training Course) | Huawei | 2025
- MATLAB Fundamentals (Training Course) | MathWorks | 2024
- HCIA-Cloud Computing (Training Course) | Huawei | 2021